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(54) DICE SPINNER

(71) Applicant: **Yarro Studios, Inc.**, Orem, UT (US)

(72) Inventor: **Tanner Yarro**, Provo, UT (US)

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(60) Provisional application No. 63/451,304, filed on Mar. 10, 2023, provisional application No. 63/455,332, filed on Mar. 29, 2023.

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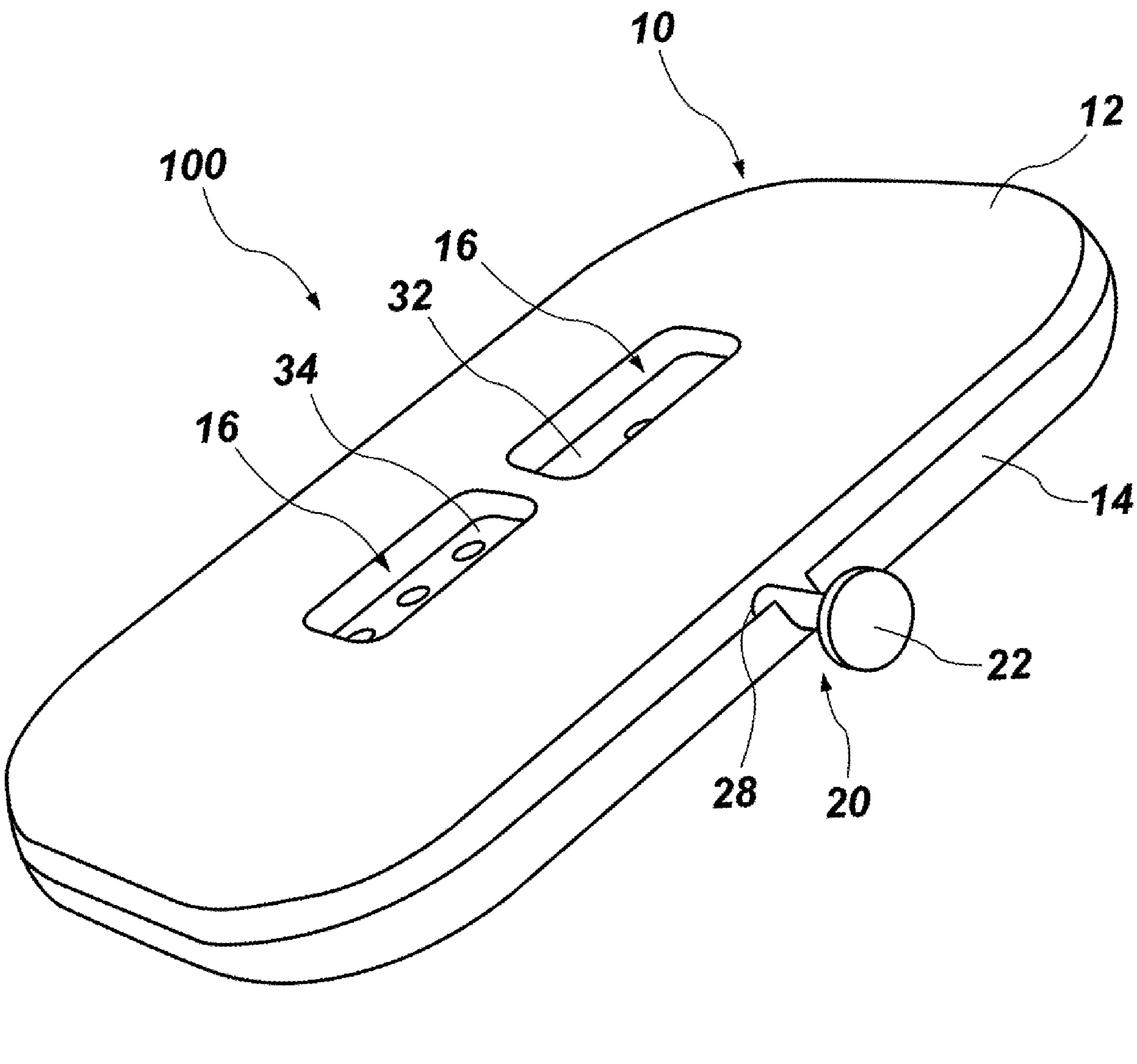
(52) **U.S. Cl.**
CPC *A63F 5/045* (2013.01)

(57) **ABSTRACT**

This disclosure relates generally to gaming equipment and/or accessories, such as dice and dice spinners. In various aspects, disclosed dice spinners include a housing having a top and a bottom, the top defining one or more windows, and a spinner housed within the housing, the spinner having a locking mechanism and at least one dice disk. Disclosed dice spinners may further include a crown in mechanical communication with the spinner such that applying pressure to the crown causes the spinner to spin and display one or more numbers through the one or more windows of the housing.



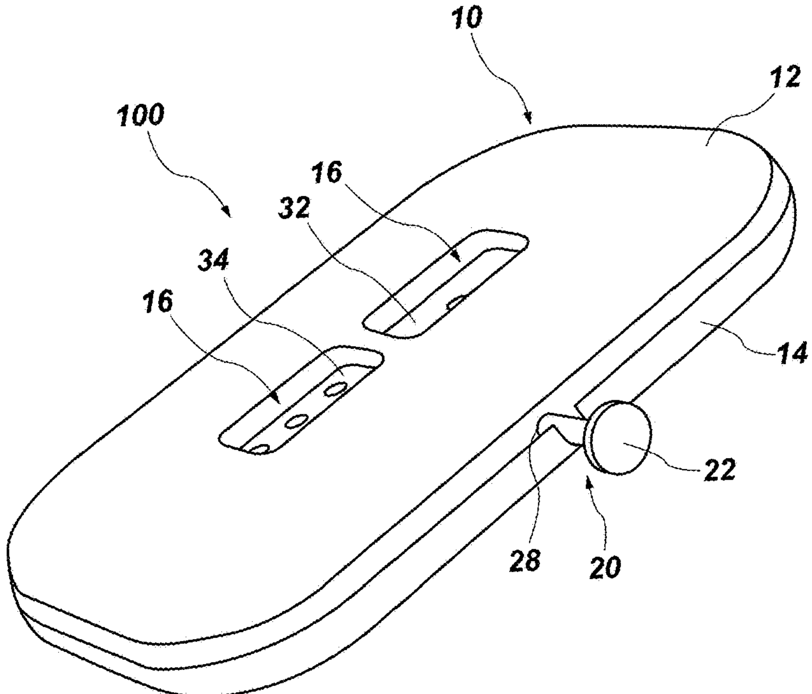


FIG. 1

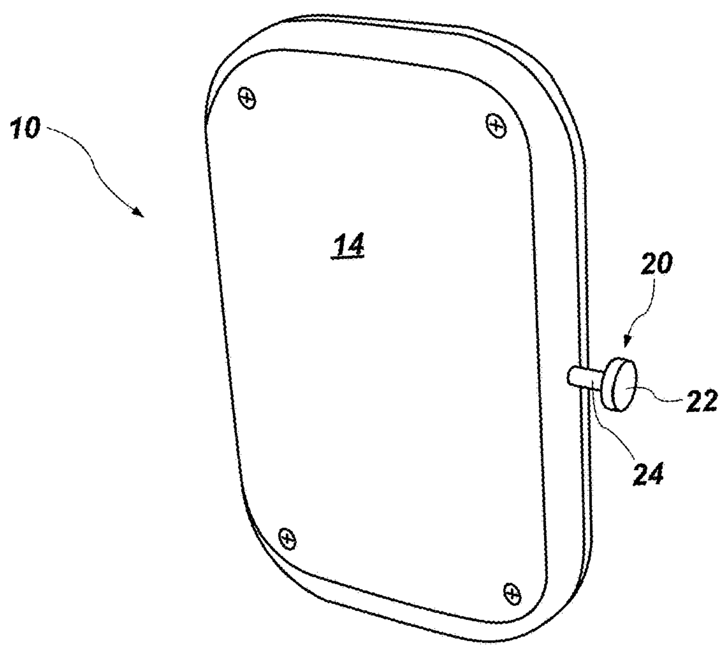


FIG. 2A

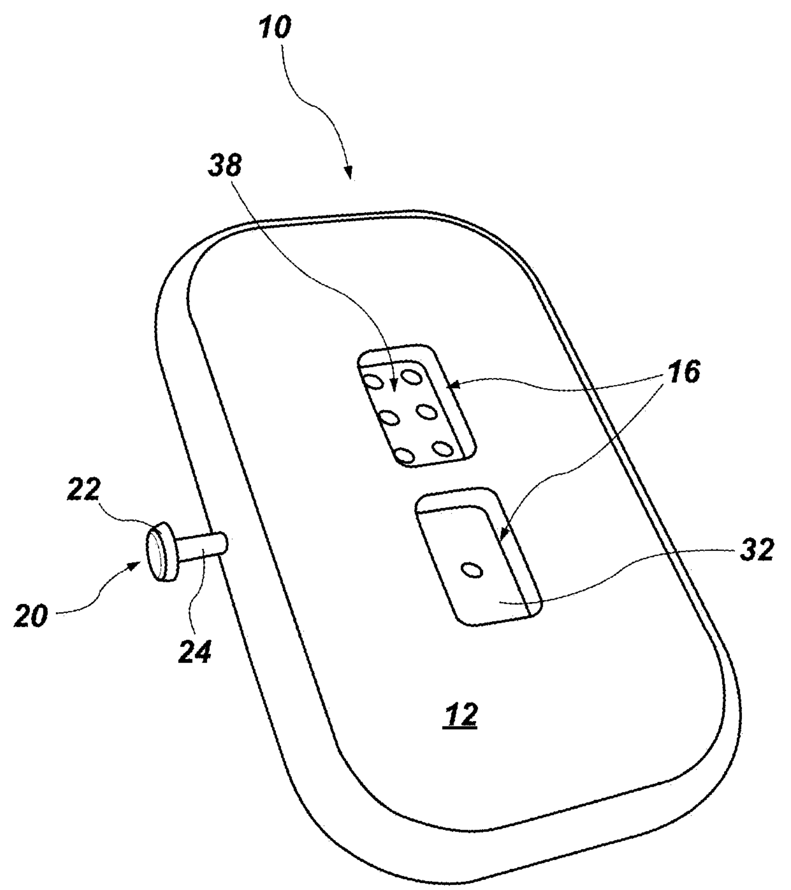


FIG. 2B

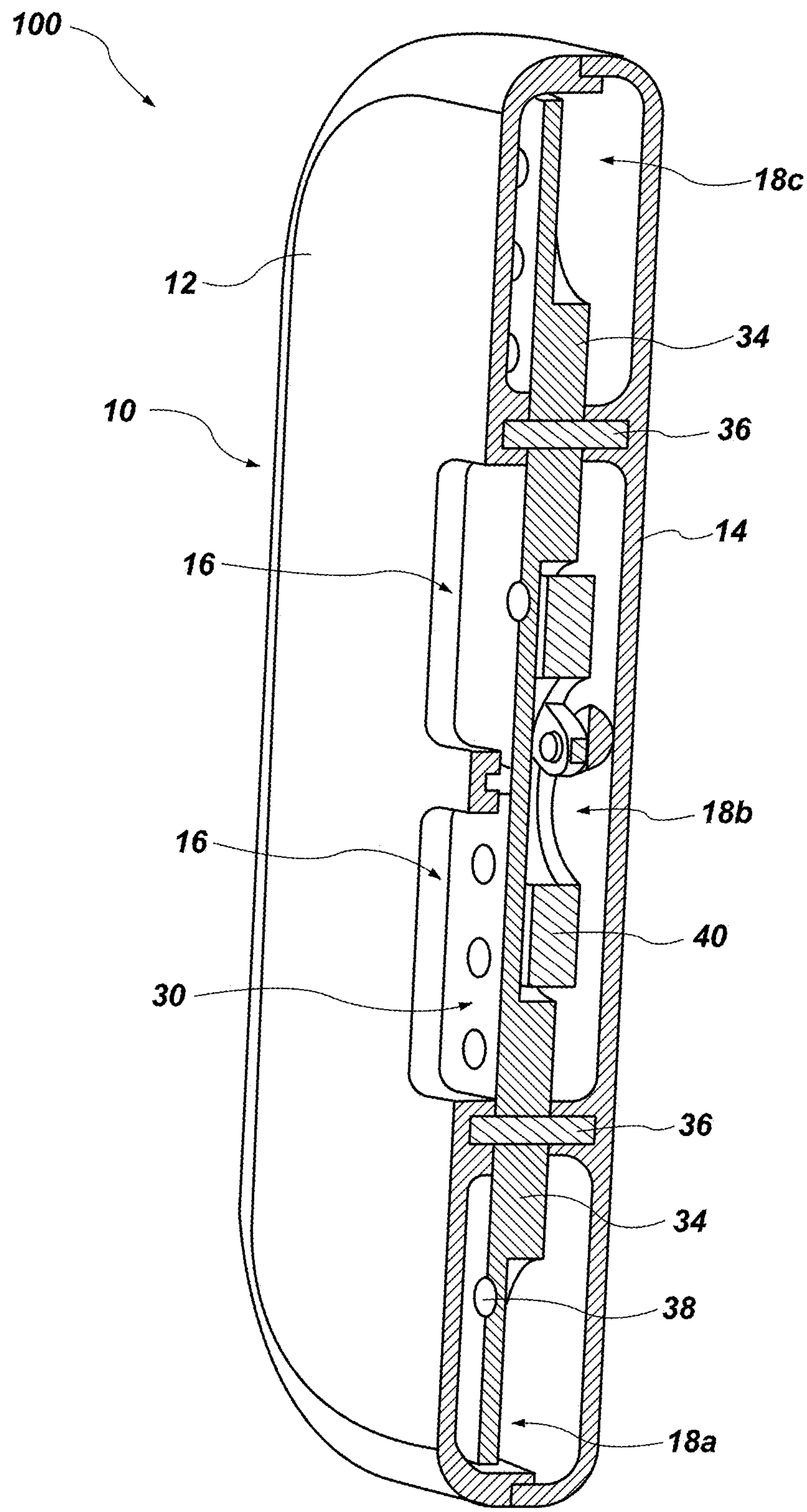


FIG. 3

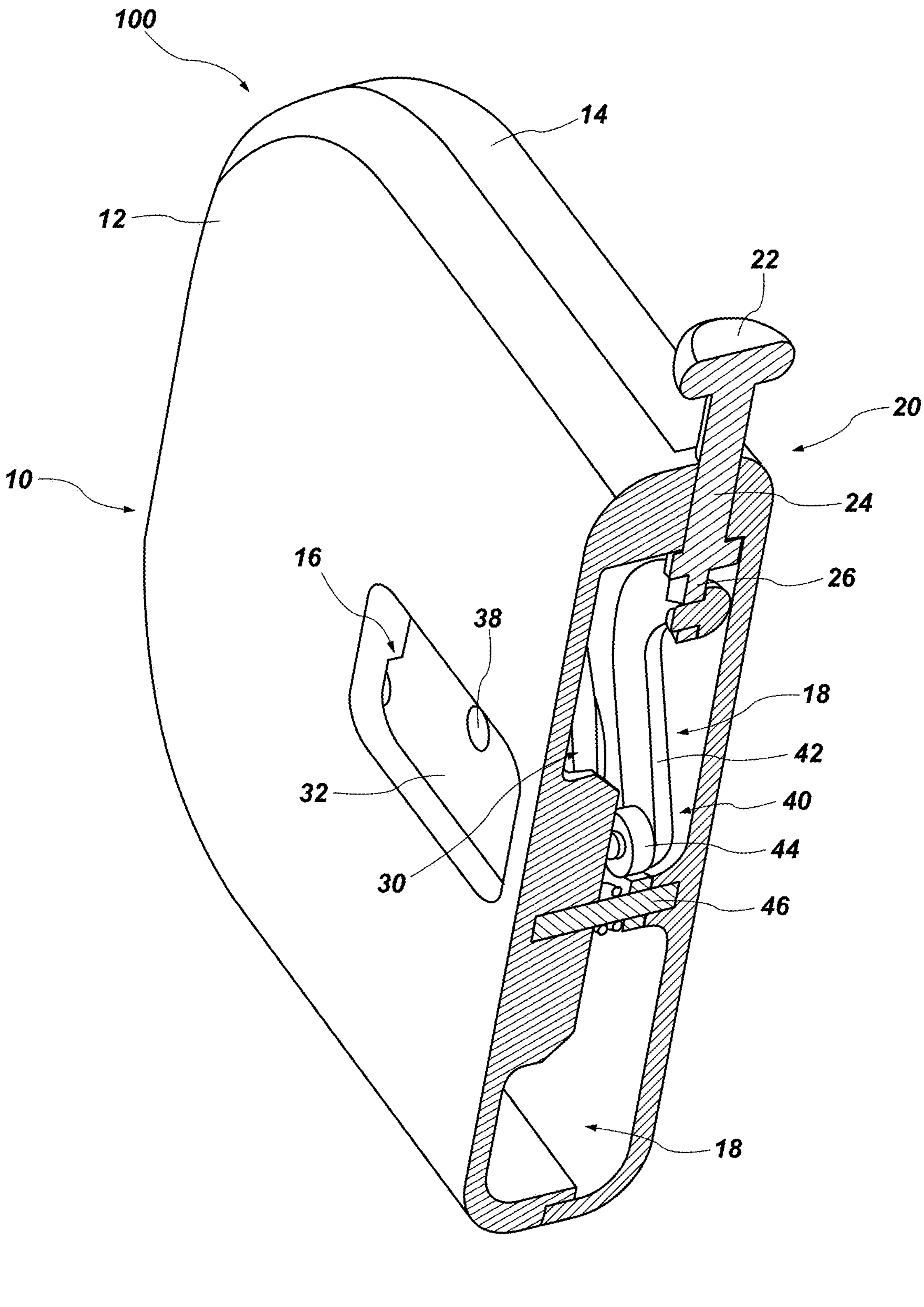


FIG. 4

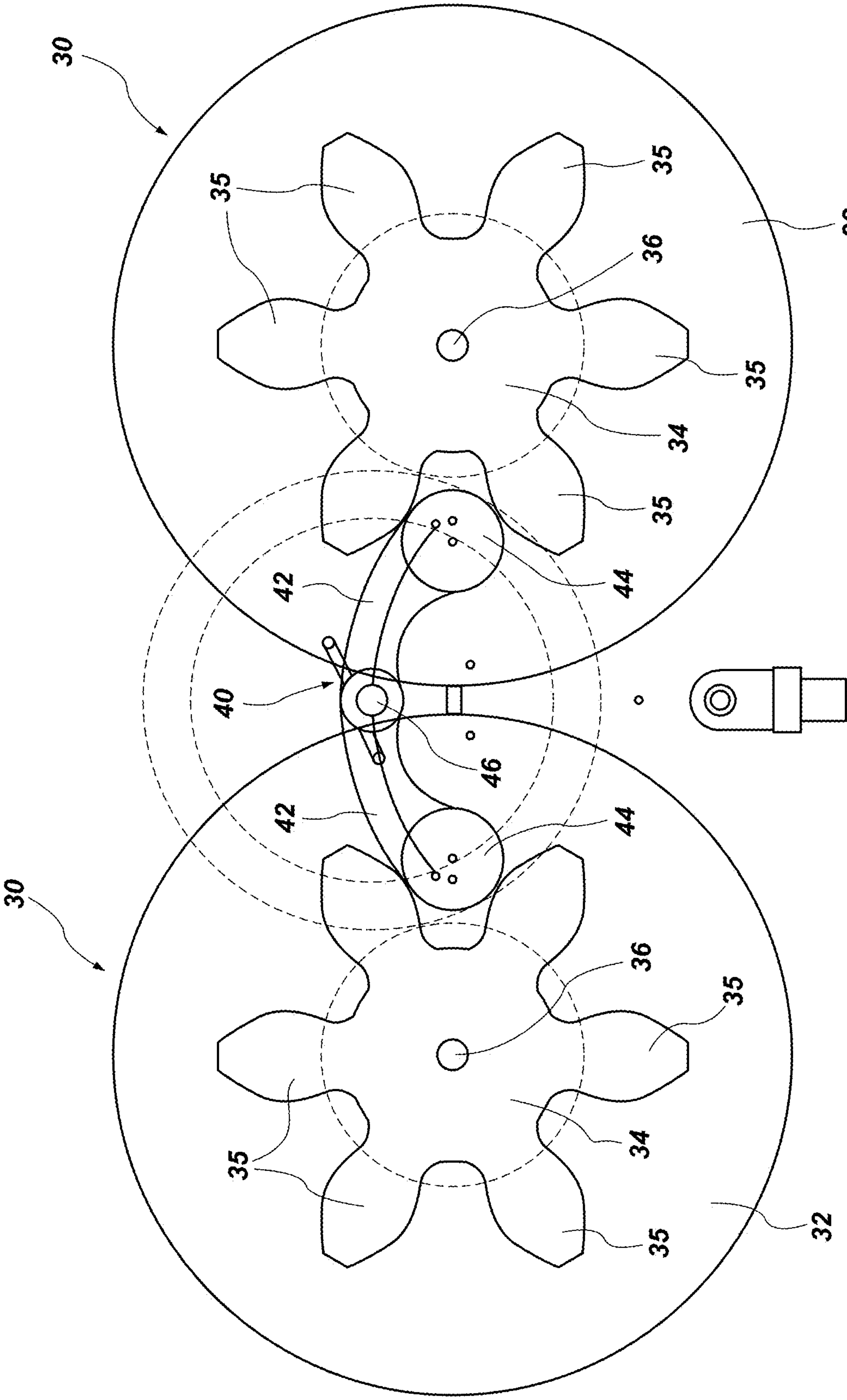


FIG. 5

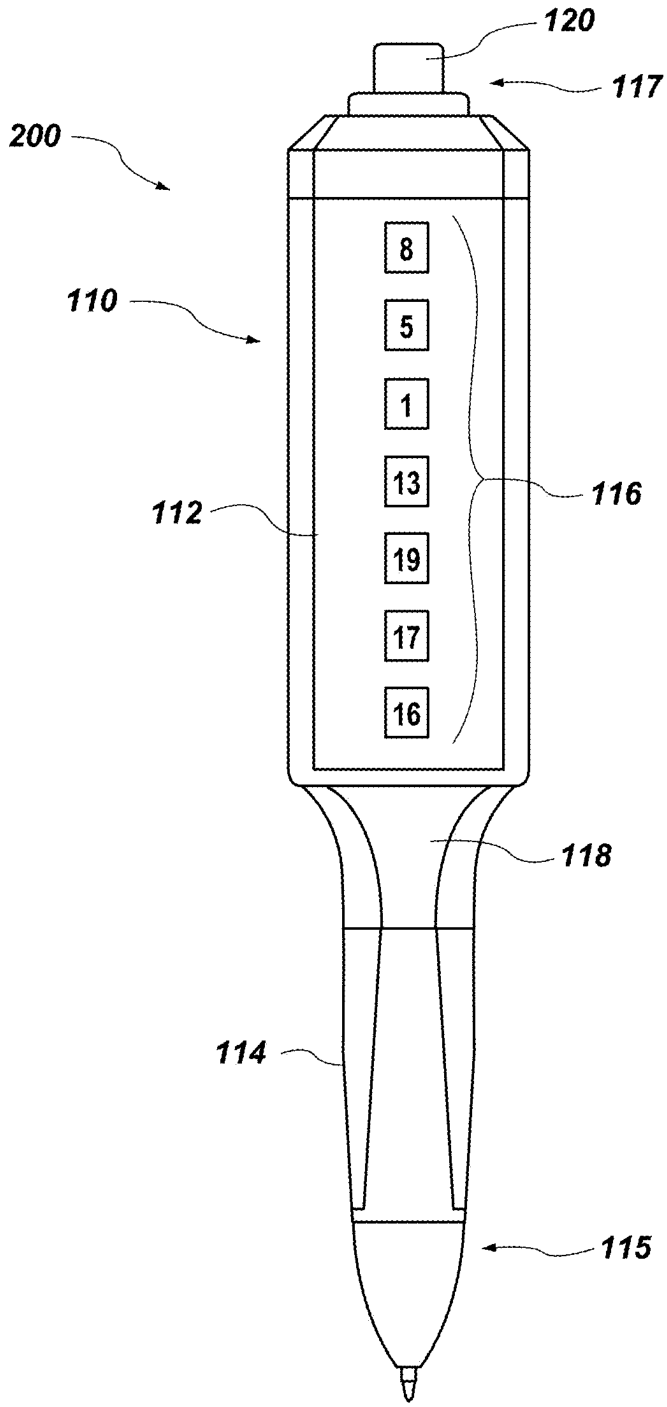


FIG. 6

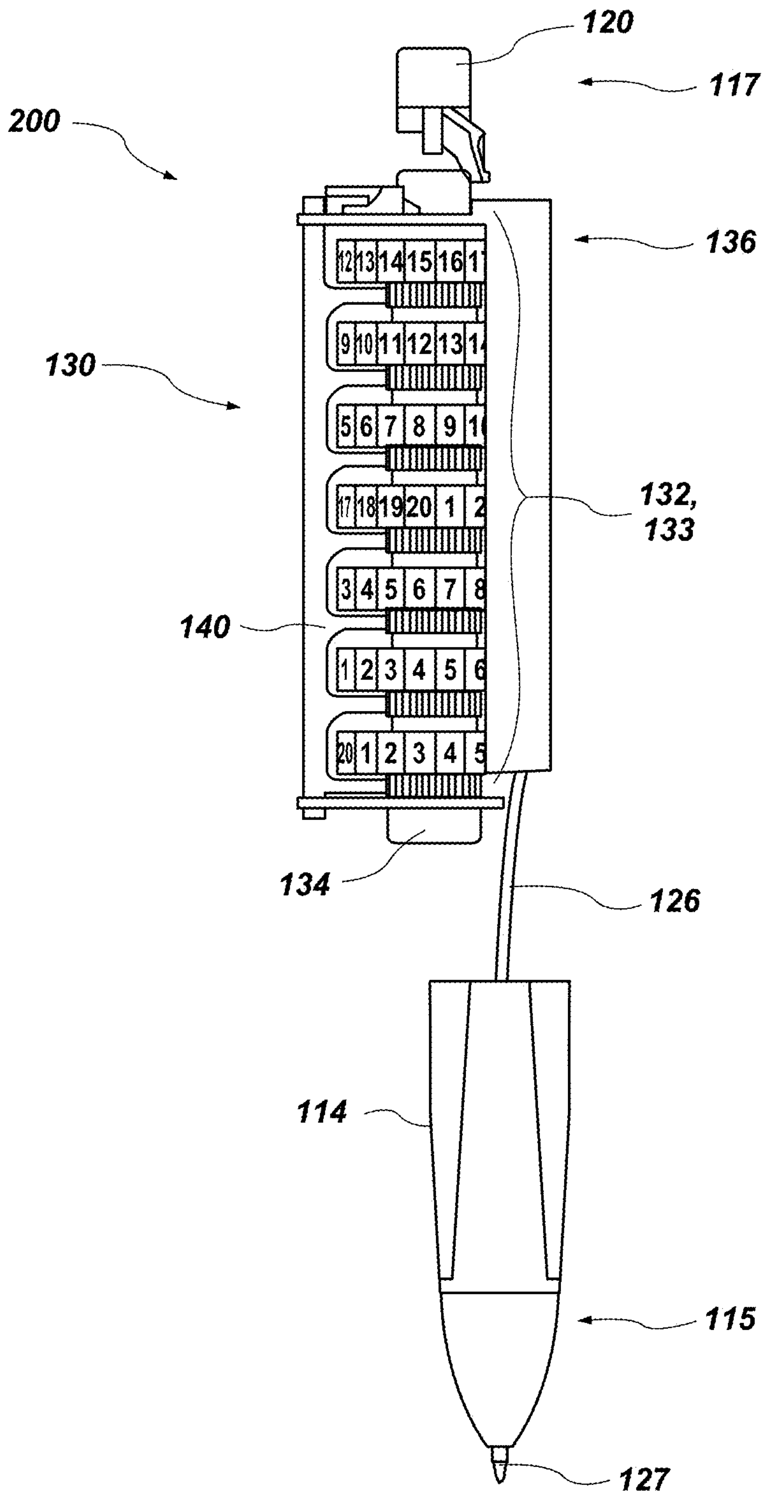
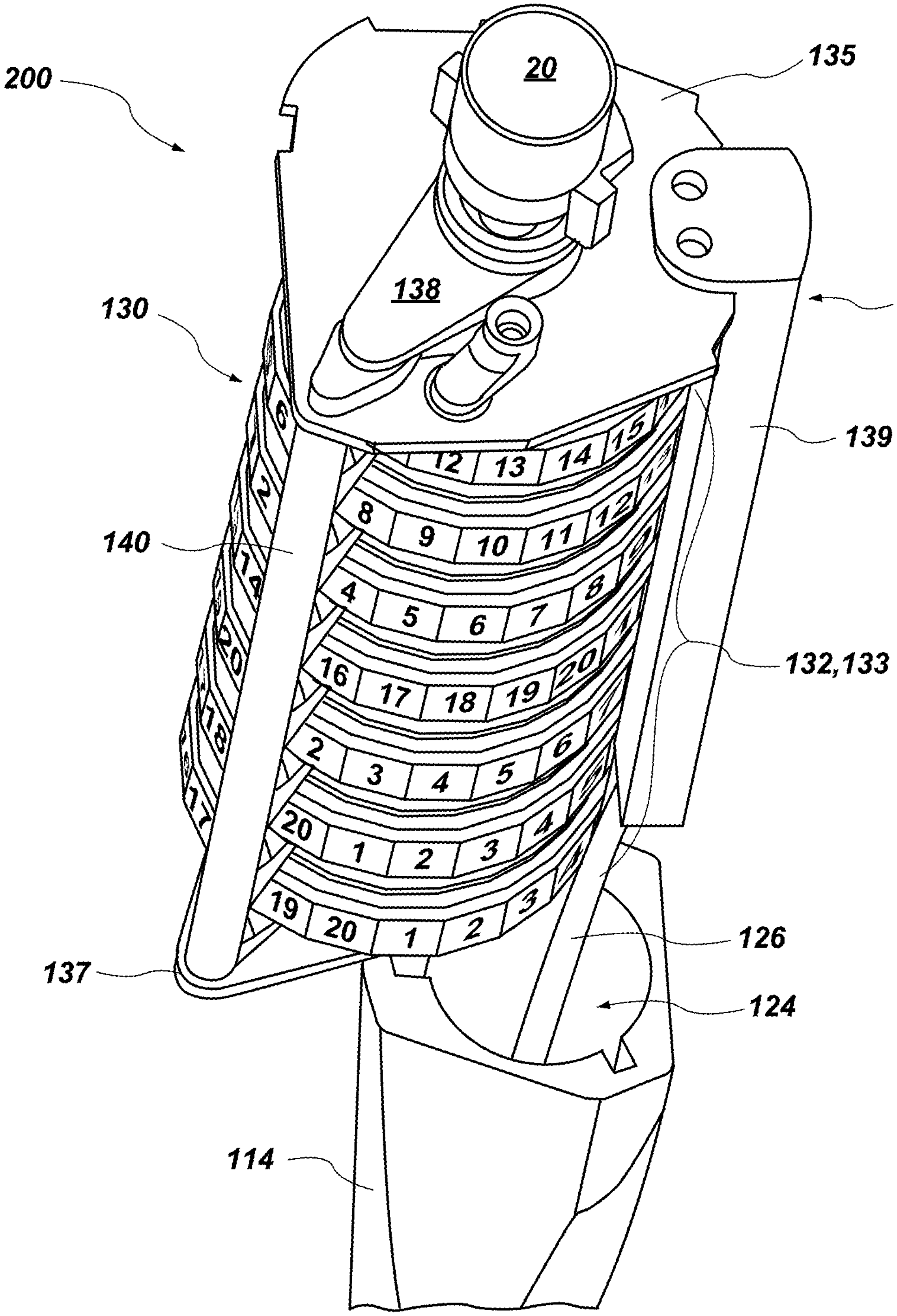


FIG. 7


200
20
135
138
136
130
139
140
132,133
126
137
124
114
12
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11
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FIG. 8

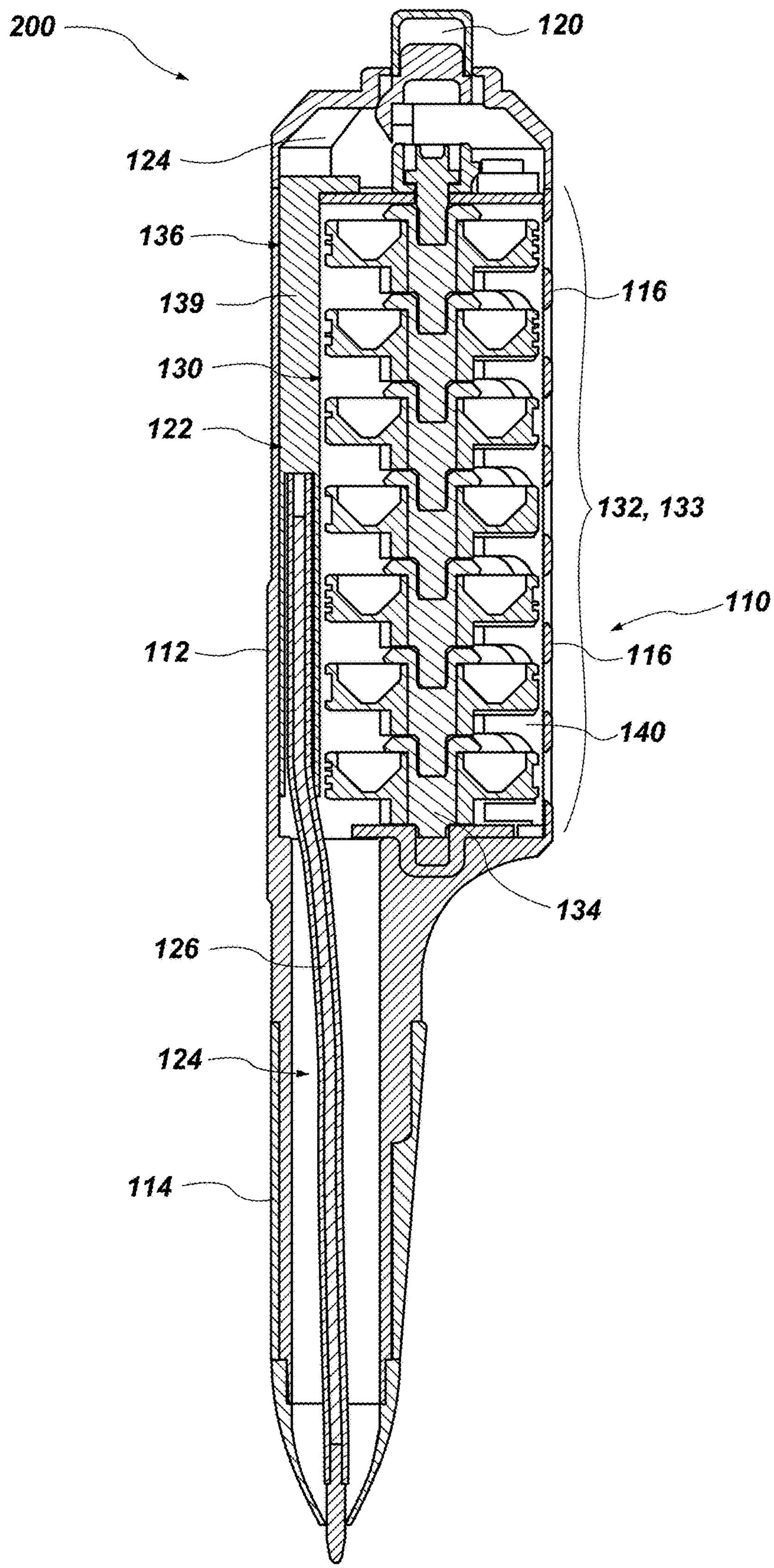
200
120
124
136
139
130
122
116
132, 133
110
112
116
140
134
126
124
114

FIG. 9

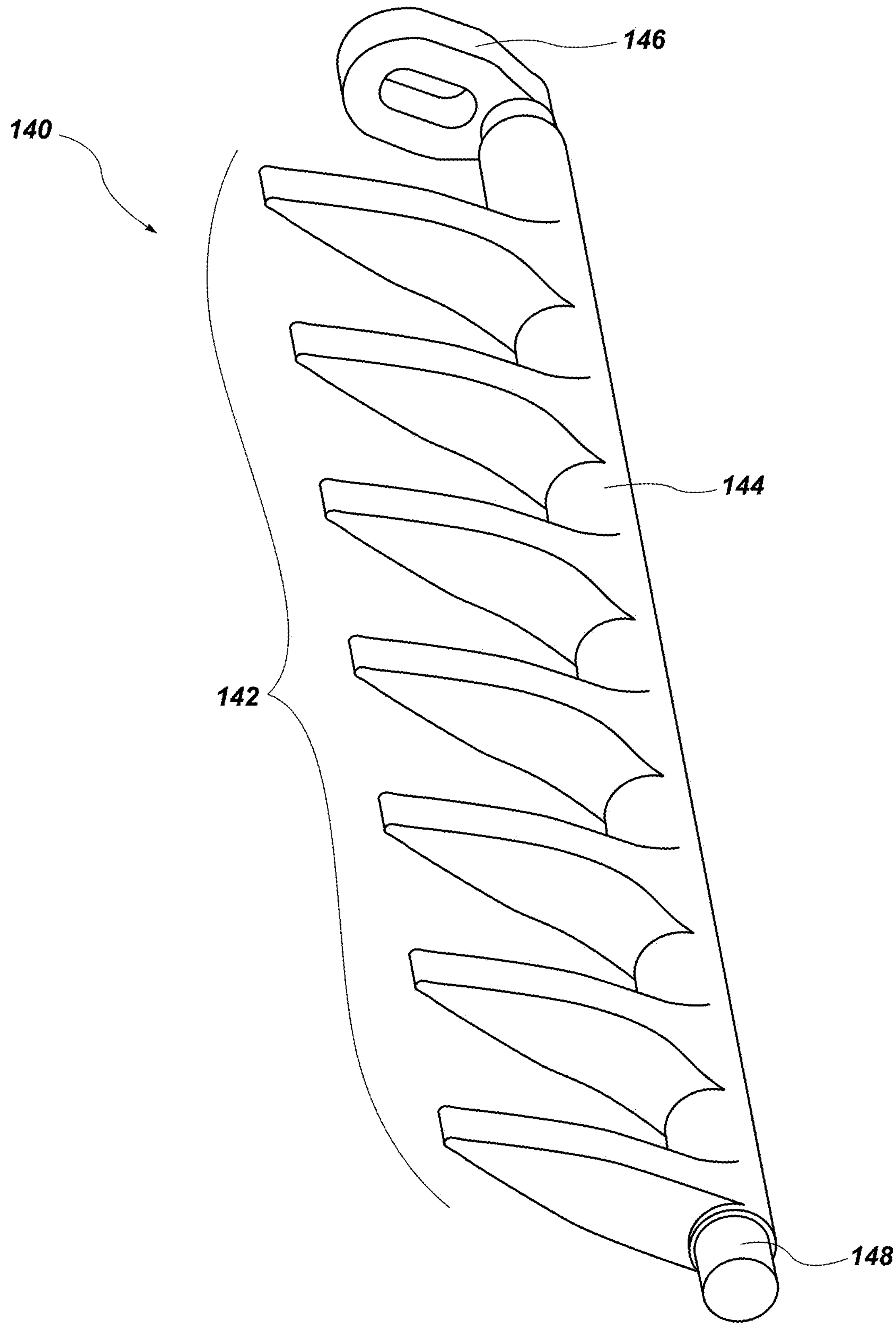

FIG. 10

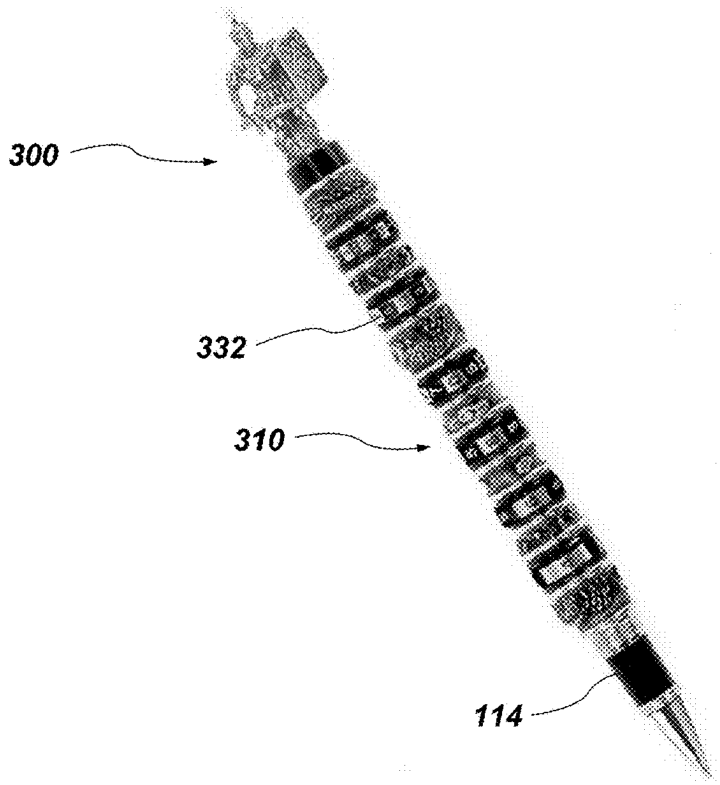

FIG. 11A

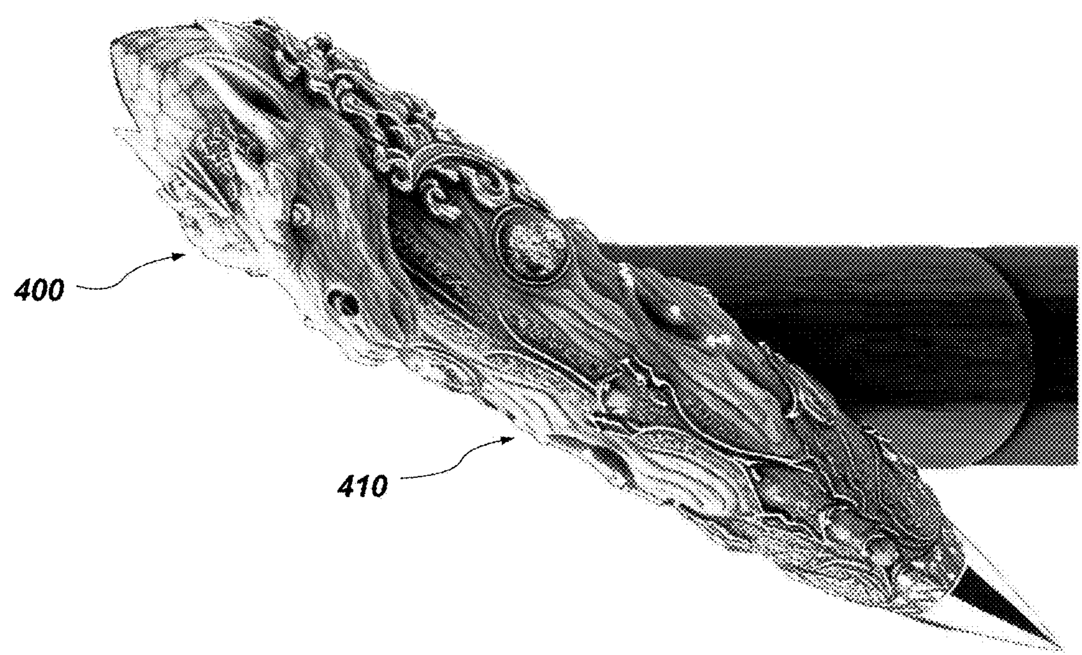

FIG. 11B

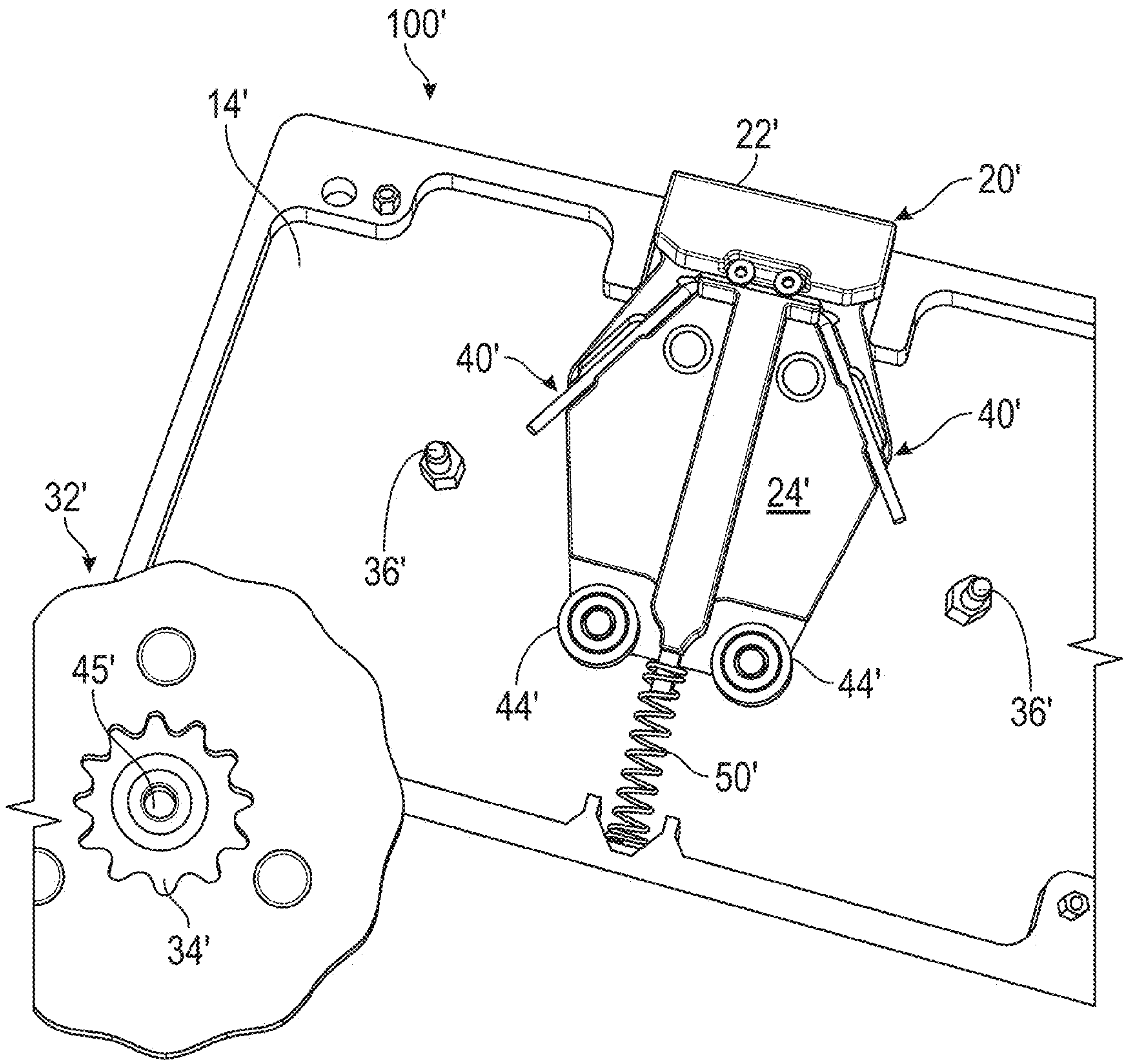

FIG. 12

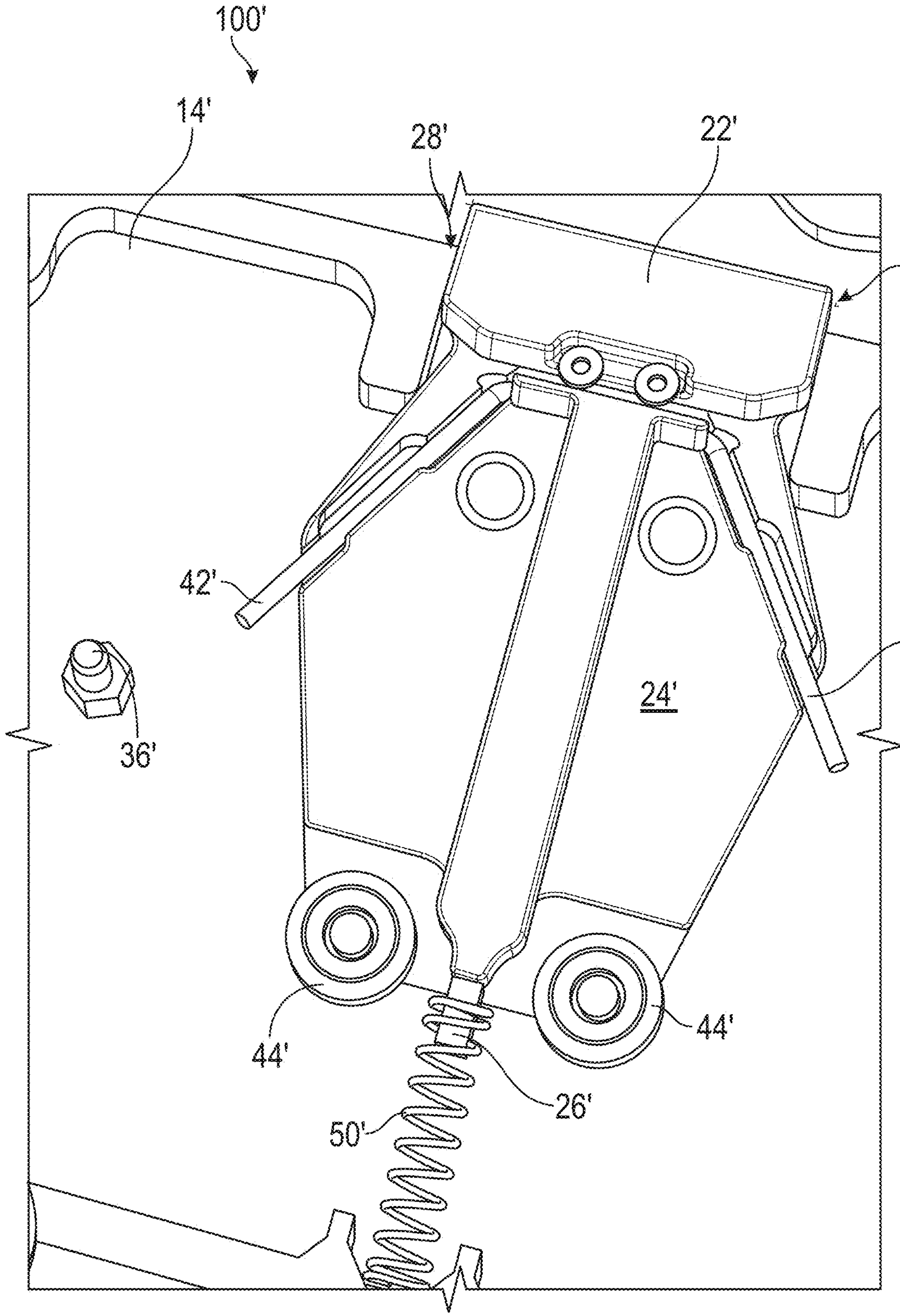

FIG. 13

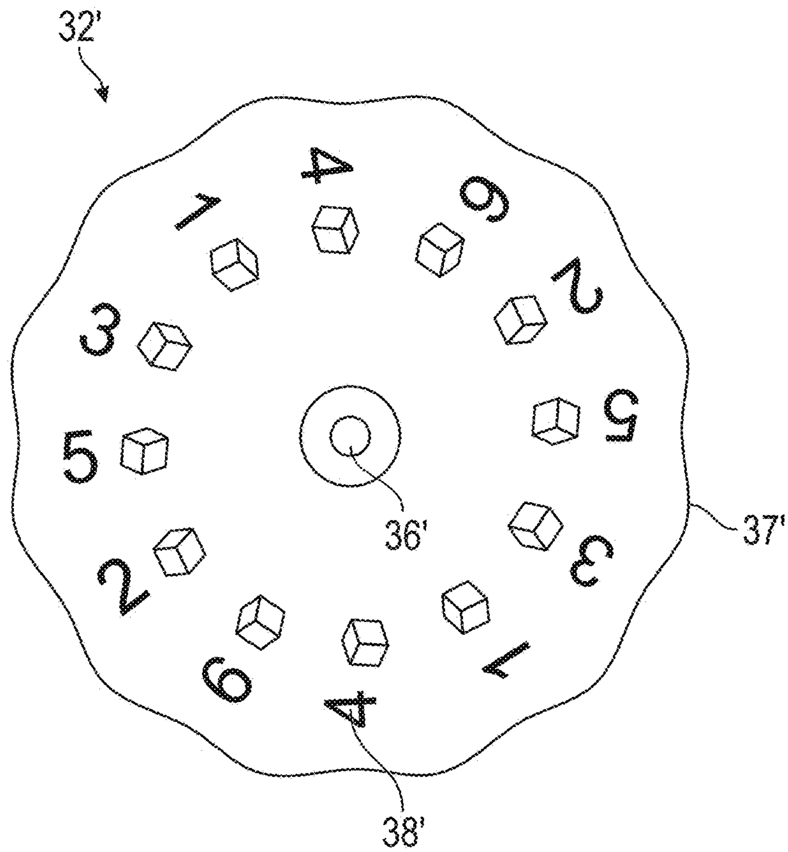

FIG. 14A

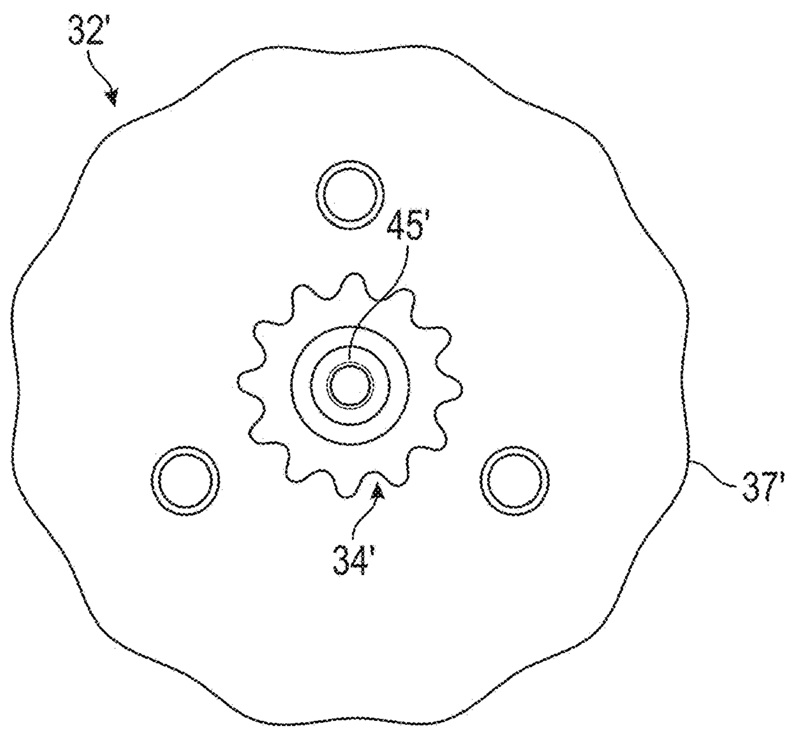

FIG. 14B

DICE SPINNER

CROSS REFERENCE TO RELATED APPLICATION

[0001] This application is a continuation-in-part of U.S. patent application Ser. No. 18/382,976 filed on Oct. 10, 2023 and titled DICE SPINNER (the "'976 Application"), which claims priority to U.S. Provisional Patent Application No. 63/451,304, filed on Mar. 10, 2023 and titled DICE SPINNER (the "'304 Application") and U.S. Provisional Patent Application No. 63/455,332, filed on Mar. 29, 2023 and titled DICE PEN (the "332 Application"). A claim to priority to the '976 Application, '304 Application, and '332 Application is hereby made pursuant to 35 U.S.C. § 119 (e). The entire disclosures of each of the '976 Application, '304 Application, and '332 Application are incorporated herein by reference.

TECHNICAL FIELD

[0002] This disclosure relates generally to games and gaming equipment, such as dice and dice spinners. More specifically, to a device for spinning a disk or a plurality of disks to be used in a game that involves chance or other game or games such as Dungeons and Dragons ("D&D"). The features disclosed herein may be numerous in nature and may be employed in various different ways to provide the same or similar results. The following description may refer to the design flat housing shaped spinner or a pen like apparatus all of which refer to the same element or mechanics in its use.

SUMMARY

[0003] Disclosed are systems, devices, and/or methods of use thereof regarding gaming equipment and/or accessories, such as dice and dice spinners. In various aspects, disclosed dice spinners include a housing having a top and a bottom, the top defining one or more windows, and a spinner housed within the housing, the spinner having a locking mechanism and at least one dice disk. Disclosed dice spinners may further include a crown in mechanical communication with the spinner such that applying pressure to the crown causes the spinner to spin and display one or more numbers through the one or more windows of the housing.

[0004] In some embodiments, disclosed dice spinners may include a housing having a top and a bottom, with the top defining one or more windows. The dice spinners may further include a first spinner housed within the housing, the first spinner including a locking mechanism, a platform disposed over a portion of the locking mechanism, and a dice disk disposed on the platform. In some embodiments, dice spinners also include a crown in mechanical communication with the locking mechanism such that applying pressure to the crown causes the locking mechanism to disengage the platform, allowing the platform and the dice disk to spin.

[0005] Other aspects of the disclosed subject matter, as well as features and advantages of various aspects of the disclosed subject matter, should be apparent to those of ordinary skill in the art through consideration of the ensuing description, the accompanying drawings, and the appended claims.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] In the drawings:

[0007] FIG. **1** illustrates a top, perspective view of a dice spinner according to the present disclosure;

[0008] FIG. **2**A illustrates a back view of the dice spinner of FIG. **1**;

[0009] FIG. **2**B illustrates a front view of the dice spinner of FIG. **1**;

[0010] FIG. **3** illustrates a cross-sectional view of the dice spinner of FIG. **1** taken through a sagittal plane;

[0011] FIG. **4** illustrates a cross-sectional view of the dice spinner of FIG. **1** taken through a transverse plane;

[0012] FIG. **5** illustrates an internal view of the dice spinner of FIG. **1**;

[0013] FIG. **6** illustrates an alternate embodiment of a dice spinner according to the present disclosure from a side view;

[0014] FIG. **7** illustrates a side view of internal components of the dice spinner of FIG. **6**;

[0015] FIG. **8** illustrates a top, perspective view of the dice spinner of FIG. **7**;

[0016] FIG. **9** illustrates a cross-sectional side view of the dice spinner of FIG. **6**;

[0017] FIG. **10** illustrates one embodiment of a locking mechanism of the dice spinner of FIG. **6**, according to the present disclosure;

[0018] FIGS. **11**A and **11**B illustrate additional embodiments of a dice spinner, which may be a pen, according to the present disclosure

[0019] FIG. **12** illustrates a partial interval view of another embodiment of a dice spinner according to the present disclosure;

[0020] FIG. **13** illustrates a close-up view of a portion of the spinner of FIG. **12**; and

[0021] FIGS. **14**A and **14**B illustrate top and bottom views, respectively, of dice disks for use with the dice spinner of FIGS. **1-9** and/or the dice spinner of FIG. **12**.

DETAILED DESCRIPTION

[0022] Humans are playful animals and, generally, enjoy playing games as part of social activities. Many games may be played with relatively few/simple components, such as a deck of cards. Some games are more intensive and require extensive equipment, such as boards, playing pieces, cards, etc. Other games utilize few components (e.g., cards), heavily incorporate elements of the imagination (e.g., setting, back story, characters, etc.), and are played on a tabletop. For example, role-playing games (RPG), such as Dungeons and Dragons® (D&D), require players to develop a backstory and abilities for their characters. The players (and their characters) then work together to accomplish a goal, go on a quest, etc.

[0023] Often, RPGs still involve game pieces to determine a player's or character's moves, lives, fire power, accuracy, etc. For example, dice are often used in D&D to determine who strikes first, how accurate the strike is, and how powerful the defense is. Dice are similarly used in games like Risk®. Dice are also often used in board games to determine how far a player can advance, among other things.

[0024] In using dice, a player typically shakes them up in a cup or their hand, and then throws the dice in the playing arena to see what numbers are generated. Importantly, dice are generally considered to generate random combinations of numbers.

[0025] Disclosed are systems, devices, and/or methods of use thereof regarding gaming equipment and/or accessories, such as dice and dice spinners which may include a long tubular like spinner or a flat planer like spinner. In various aspects, disclosed dice spinners include a housing defining one or more windows, and a spinner housed within the housing, the spinner having a locking mechanism and at least one dice disk. In some embodiments, the at least one dice disk includes or bears indicia, corresponding to a numerical value. Disclosed dice spinners may further include a crown, or button, in mechanical communication with the spinner such that applying pressure to the crown causes the spinner, or spinners, to spin. Upon release of the crown, the spinner may stop spinning and display one or more numbers through the one or more windows of the housing. Alternatively upon release of the crown the spinner may continue spinning until it comes to a stop on its own. Additionally, the spinner may include a locking mechanism and a plurality of wheels or disks. Additionally, disclosed spinners may include a clicker disposed on a top, or proximal, surface of the body. The clicker may be in communication with the spinner such that applying pressure to the clicker releases the locking mechanism, allowing the spinner to spin.

[0026] In some embodiments, disclosed dice spinners may include a housing having a top portion and a bottom portion, with the top defining one or more windows. The dice spinners may further include a first spinner housed within the housing, the first spinner including: a locking mechanism, a platform disposed over a portion of the locking mechanism, and a dice disk disposed on the platform. One or more of the locking mechanism and the platform may be secured to an interior of the housing. In some embodiments, dice spinners also include a crown in mechanical communication with the locking mechanism such that applying pressure to the crown causes the locking mechanism to disengage the platform, allowing the platform and the dice disk to spin, or alternatively disengaging the platform and pushing, or forcing the dice disk to spin.

[0027] FIG. **1** illustrates a top, perspective view of a dice spinner **100** according to the present disclosure. As illustrated, the dice spinner **100** includes a housing **10**, where the housing has a top portion **12** and a bottom portion **14**. The top portion **12** is secured to the bottom portion **14** via screws, clips, snaps, adhesives, welding, or other appropriate securement mechanism. In some embodiments, the top portion **12** is removably secured to the bottom portion **14**. The top portion **12** of the housing **10** may define one or more windows **16**, such as the two windows **16** illustrated. In some embodiments, the windows **16** allow a user of the dice spinner **100** to visually see dots, circles, other indicia, etc. corresponding to numerical values. For example, a portion of disks **32**, or dice disks, are visible through the one or more windows **16**, where the dice disks **32** carry or bear indicia corresponding to numerical, or other, values. In this way, the dice spinner **100** can approximate or mimic a roll of dice during, for example, game play.

[0028] In some embodiments, the dice spinner **100** additionally includes a crown **20**, or button. The crown **20** may be received or disposed within a void **28** defined by the top and bottom portions **12**, **14** of the dice spinner **100**. In some embodiments, the crown **20** has a head **22** and a body **24** that protrudes through the void **28** into an interior of the housing **10**. As discussed more fully below, the crown **20** interfaces or mechanically engages with a locking mechanism, and/or spinning mechanism, housed within the interior of the housing **10**.

[0029] FIG. **2**A illustrates a back view and FIG. **2**B illustrates a front view of the dice spinner **100** of FIG. **1**. As seen in FIG. **2**A, the bottom or back portion **14** of the housing **10** may receive a plurality of screws or fasteners to secure the bottom portion **14** to the top portion **12** of the housing **10**. The fasteners may reversibly secure the top portion and the bottom portion allowing a user to open the housing as needed to interchange the disks **32** as desired by a user. As seen in FIG. **2**B, the top portion **12** defines one or more windows **16**, through which can be seen at least a portion of the dice disks **32**. As illustrated, the dice disks **32** carry dots or indicia **38** corresponding to numerical, or other game, values, in this case a one (1) and a six (6). The head **22** and body **24** of the crown **20** extend a distance from the housing **10**, such that a user can manipulate the crown **20** to cause the dice disks **34** housed within the housing **10** to spin and display new numerical values. The crown **20** may be "pushed" or "twisted" in a direction to articulate the spinners and/or locking mechanism.

[0030] FIG. **3** illustrates a cross-sectional view of a dice spinner **100** taken through a sagittal plane and FIG. **4** illustrates a cross-sectional view of the dice spinner **100** taken through a transverse plane. As illustrated, the top and bottom portions **12**, **14** of the housing **10** defines a plurality of inner cavities **18**. As illustrated in FIG. **3**, the plurality of inner cavities **18**, or voids, includes three (3) cavities **18**. The three cavities **18** may be one large cavity or may be divided into a plurality of cavities. The plurality of cavities **18** house and/or receive a myriad of elements of the dice spinner **100**. For example, a middle or center cavity **18***b* houses the locking mechanism **40** and a portion of a first spinner **30** wherein the locking mechanism **40** may include a first and second arm that may engage the first spinner **30** and a second spinner **30***a* respectively. A first cavity **18***a* may reside toward a left side of the dice spinner **100** when the dice spinner has the top portion **12**, or front portion, facing up. A third cavity **18***c* may reside toward a right side of the dice spinner **100** when the dice spinner has the top portion **12** facing up. As discussed more fully below with respect to FIG. **5**, the spinner **30** may include a dice disk **32**. The dice disk **32** can bear indicia **38** corresponding to one or more numerical values, where the indicia **38** is visible through the one or more windows **16** defined in the top half **12** of the housing **10**. Disk pivot or anchor points **36**, which may be pins, may define borders of the plurality of cavities **18**. The pivot points **36** may also be the point about which dice disks **32** and/or the locking mechanism **40** pivot and/or spin or rotate around the disk pivot points **36**, or disk pin. In some embodiments, the locking mechanism **40** pivots about a locking pivot point **46**.

[0031] As illustrated in FIG. **4**, a portion of the crown **20** extends into the housing **10**. Specifically, in some embodiments, a portion of the body **24** is sandwiched between the top and bottom halves **12**, **14** of the housing **10**. An end **26** of the crown **20** may interface and/or mechanically engage the locking mechanism **40**. The end **26** may interface with a body **42** of the locking mechanism **40** (see FIG. **5**). In some embodiments, manipulation of the crown **20** causes a stopper **44** of the locking mechanism **40** to release from a dice disk **32**. For example, pressure may be applied to the head **22** of the crown **20**, causing the body **24** and end **26** of the crown **20** to extend further into the housing **10**. Such extension may cause the stopper **44** to disengage and/or release the dice disk **32**. Upon release of the stopper **44**, the dice disk **32** will spin around a pivot point **36** (of FIG. **3**). When the crown **20** is released, the body **24** and end **26** of the crown **20** will retract and extend away from the housing **10**. Upon release of the crown **20**, the stopper **44** of the locking mechanism **40** will re-engage with the dice disk **32**, causing the dice disk **32** to stop spinning. When the dice disk **32** stops spinning, indicia corresponding to a numerical value will be visible through the one or more windows **16**.

[0032] In some embodiments, manipulation of the crown **20** may include twisting instead of applying pressure to the crown **20**. Upon twisting the crown **20**, the end **26** will disengage the locking mechanism **40**. Disengagement of the locking mechanism **40** by the crown **20** causes the locking mechanism **40** to release and disengage the dice disk **32**. Upon release of the locking mechanism **40**, the dice disk **32** will spin around a pivot point **36**. When the crown **20** is twisted again (e.g., in the opposite direction), the end **26** of the crown **20** will re-engage the locking mechanism **40**. Upon ren-engagement of the locking mechanism **40** by the crown **20**, the locking mechanism **40** will re-engage with the dice disk **32**, causing the dice disk **32** to stop spinning. When the dice disk **32** stops spinning, indicia corresponding to a numerical value will be visible through the one or more windows **16**.

[0033] FIG. **5** illustrates one embodiment of a spinner **30** according to the present disclosure. In some embodiments, the spinner **30** includes a sprocket **34**, or cog and the dice disk **32**. The sprocket **32** may be positioned with the housing engaging disk pin **36** on an interior of the housing **10**, such as on an interior of the bottom half **14** of the housing **10** (see FIGS. **1-2**B). The dice disk **32** is disposed about the sprocket **34** and secured via a pivot point **36**. The sprocket **32** may be adhered to or fastened to the dice disk **32**. In some embodiments, the spinner **30** additionally includes a locking mechanism **40**. In some embodiments, the locking mechanism **40** is separate from the spinner **30**. The locking mechanism **40** may include a body **42** and a stopper **44**, where the stopper **44** engages or otherwise interfaces with the dice disk **32**, ultimately engaging the sprocket **34**. The locking mechanism **40** may be secured to the housing **10** via the locking pivot point **46**. The locking mechanism **40** may pivot about the locking pivot point **46**.

[0034] The dice disks **32** may include a plurality of wings **35**, or arms, or recesses and valleys, where each wing **35** of the plurality may carry or bear indicia corresponding to a numerical value, or between each wing, or in between each wing, bears the indicia corresponding to the numerical value. Each wing **35** may be a tooth in the sprocket **34**. In some embodiments, the plurality of wings **35** may include an odd number of wings. In some embodiments, the plurality of wings **35** may include an even number of wings. In some embodiments, the housing **10** may house more than one spinner **30** and the dice disks **32** of each spinner **30** may include the same number of wings **35** in the plurality. Alternatively, the dice disks **32** of each spinner **30** may include a different number of wings **35** in the plurality. Any number of wings **35**, or teeth, may be utilized for any of a plurality of indicia or numerical values and thus the stopper **44** may be of any appropriate size and shape to engage the sprocket **34** to spin and lock the disks **32**.

[0035] Each of the wings **36** may include a valley **37**, between two wings **36**, that engages the stopper **44** of the locking mechanism **40** to stop the disks **32** at a position that correlates with the indicia or numerical value.

[0036] In some embodiments, the wings **35** bear sequential indicia. Alternatively, the wings **35** may bear non-sequential and/or random indicia. In embodiments where the housing **10** houses two spinners **30**, the wings **35** of each spinner **30** may bear the same indicia (sequential, non-sequential, random, etc.). In other embodiments where the housing **10** houses two spinners **30**, the wings **35** of each spinner **30** may bear different indicia (e.g., different numerical values, sequential and non-sequential values, etc.).

[0037] As before, when the crown **20** is engaged (i.e., upon an application of pressure or other manipulation of the crown **20**), the crown **20** causes the locking mechanism **40** to release from the dice disk **32**. Specifically, manipulation of the crown **20** causes the body **42** of the locking mechanism **40** to pivot about locking pivot point **46**, or locking pin, such that the stopper **44** is released from the dice disk **32**. The dice disk **32** then spins about the disk pivot point **36**. When the crown **20** is disengaged, or released, the locking mechanism **40** re-engages the dice disk **32**, causing the dice disk **32** to stop spinning. When the dice disk **32** stops spinning, indicia **38** corresponding to a numerical value will be displayed through the one or more windows **16** defined in the top half **12** of the housing **10**. Specifically, a portion of at least one disk **32**, and the carried indicia, will be visible through the one or more windows **16**.

[0038] In some embodiments, the dice spinner **100** houses one or more spinners **30**, such as two spinners **30** as illustrated in FIG. **5**. In some embodiments, the locking mechanism **40** includes two opposing bodies **42** as a unitary piece, joined at the locking pivot point **46**. In some embodiments, an end of each body **42** carries a stopper **44** for engagement with a dice disk **32** and/or a sprocket **34** of the spinner **30**. In some embodiments, the stopper **44** may be a rubber or silicon stopper. In some embodiments, twisting the crown **20** clockwise causes the locking mechanism **40** to release or disengage a first spinner **30** but not a second spinner **30**. In some embodiments, twisting the crown **20** counterclockwise causes the locking mechanism **40** to release or disengage the second spinner **30** but not the first spinner **30**. It will be further appreciated that a plurality of disks **34** are considered and thus a plurality of stoppers **44** as well as locking mechanisms **40** to engaged and disengage those discs is contemplated. In the present disclosure a single actuator **20** may be utilized or a plurality of actuators which may engage one disk **32** or a plurality of disks **34**.

[0039] It will be appreciated that the spinners **30** may be in constant tension and will spin when the locking mechanism **40** is released via the crown **20**. In the alternate, the spinners may be at rest and "spin" when the crown **20** is actuated "pushing" the spinners **30** in either direction, clockwise or counterclockwise, and then stopped with the crown **20** is actuated again to engage the spinners **30** as a stopper.

[0040] Additionally, a spinner may include a more vertical integration as set forth in FIGS. **6-11**. In this configuration a dice spinner **200** may reside within a pen but it will be appreciated that the pen is in accessory to the dice spinner **200**. Disclosed are systems, devices, and/or methods of use thereof relating to a dice spinner **200**. In various aspects, these disclosed dice spinners include a body having a grip portion and a dice portion, where the dice portion defines an internal cavity and a plurality of exterior windows. The spinner may include a locking mechanism and a plurality of wheels. Additionally, disclosed dice spinner may include a clicker disposed on a top, or proximal, surface of the body. The clicker may be in communication with the spinner such that applying pressure to the clicker releases the locking mechanism, allowing the spinner to spin.

[0041] In some embodiments, disclosed dice spinners may include a body defining a first internal cavity and a plurality of external windows. The dice spinners may also include a spinner disposed and housed within the internal cavity, and a clicker disposed on a top portion of the body. The clicker may be in communication with the spinner such that applying pressure to the clicker releases a locking mechanism, allowing the spinner to spin.

[0042] FIG. **7** illustrates one embodiment of a dice spinner **200** according to the present disclosure. As illustrated, the dice spinner **200** includes a body **110** having a dice portion **121** and a grip portion **114**. In some embodiments, the grip portion **14** may include a grip or other padding for holding the dice spinner **200**. The body **110** may also include a transition portion **118**, or intermediate portion, between the dice portion **112** and the grip portion **114**. The spinner **200** may additionally include a clicker **120**, or button or actuator, disposed at a top or proximal end **117** of the dice spinner **200**. The dice portion **112** of the body **110** defines an internal cavity **124** (see FIG. **4**) and a plurality of windows **116**. The plurality of windows **16** may be displayed linearly or nonlinearly. The windows **116** may be positioned on a singled side or multiple sides of the dice portion **121**. In some embodiments, one or more (e.g., a plurality of) numbers may be visible through the plurality of windows **116**. The one or more numbers visible through the plurality of windows **116** may approximate or mimic the results of a dice roll.

[0043] FIGS. **2** and **3** illustrate internal components of the dice spinner **200** of FIG. **1**. As illustrated, the internal components include a spinner **130**, spinner housing **136**, a locking mechanism **140**, and an ink cartridge **126** in the case of a pen being utilized. As discussed more fully below, the spinner **130**, spinner housing **136**, and the locking mechanism **140** are each housed or contained within the internal cavity **124** defined by the dice portion **112** of the body **110**. The clicker **120** may be in communication with the locking mechanism **140** such that applied pressure on the clicker **120** causes the locking mechanism **140** to disengage from the spinner **130**. One method, or a first method, of spinning the spinner **130** may be disengagement of the locking mechanism **140** causing or allowing the plurality of disks **132**, or wheels, of the spinner **130** to spin. In this first method the wheels **132** and/or spinner **130** may be at tension, or constant tension such that when the locking mechanism **140** is released the spinner **130** and/or wheels **132** spin until the locking mechanism **140** is reengaged. Release of the clicker **120** (i.e., removal of applied pressure) re-engages the locking mechanism **140** with the spinner **130**, such that the plurality of wheels **132** stop spinning. When the plurality of wheels **132** stop spinning, individual symbols **133**, which may be numbers, will be visible through the plurality of windows **116** on the body **110**. As before, the visible numbers approximate or mimic the results of a dice roll.

[0044] Alternatively, another method, or a second method, of spinning may be disengagement of the locking mechanism **140** and force applied to the wheels **132** and/or wheel spinner **130** until such time as the locking mechanism **140** is reengaged. In this second method the wheels **132** and/or spinner **130** may be at rest until acted upon by the actuator, or clicker **120**. As another alternative, the spinner **130** and/or wheels **132** may spin until stopped through simple slowing down and stopping without engagement of the locking mechanism **140**.

[0045] Similar to other embodiments disclosed herein, the disks **130**, may include a sprocket, or wings or teeth that allows for engagement and disengagement of the locking mechanism **140**.

[0046] It will be appreciated also that in any embodiment the clicker **120** may either need to be held or single click may unlock and spin, or allow to spin, the spinner **130** and/or wheels **132** with another click of the clicker **120** to reengage the locking mechanism. It is also contemplated herein that depending on the number of "dice" in the body of the dice portion **112** that individual locking mechanisms may be engaged and or disengaged with the clicker **120**. Thus allowing for a specified number of clicks to individually role one die, or one wheel **132**, or a plurality of dice, or wheels **132**. In the alternate a mechanism is contemplated, both internal and/or external to the housing **110**, that allows for locking of one wheel **132** or a plurality of wheels **132**, such that when force is applied to the clicker **120** only those "unlocked" die or dice will spin.

[0047] Referring to FIG. **3**, the spinner housing **136** includes a top surface **135**, a bottom surface **137**, a center post **134** wherein each of the disks **130** may rest on top of one another about the center post **134**, and a side column **139**. An anchor **138** may be disposed on the top surface **135** and may place the locking mechanism **140** in communication (e.g., mechanical) with the clicker **120**. For example, referring briefly to FIG. **5**, the locking mechanism **140** may include a top end **146** defining a hole or void. The anchor **138** may engage with the hole or void in the top end **146**, thereby securing the locking mechanism **140** to the spinner **130** and/or the spinner housing **36**. In some embodiments, the locking mechanism **140** is part of the spinner housing **136**. In some embodiments, the center post **134** is disposed through a center of each of the plurality of wheels **132** much like a pin of the previous disclosed embodiment. When the plurality of wheels **132** spin, they spin around a longitudinal axis of the center post **134**.

[0048] In the instance where the dice spinner **200** may be a pen, in some embodiments, the ink cartridge **126** (or a portion of the ink cartridge **126**) is housed within a second internal cavity **124** defined by the narrow or grip portion **114**. In some embodiments, the side column **139** defines a cavity to receive a portion of the ink cartridge **126**. In some embodiments, reception of a portion of the ink cartridge **126** within the side column **139** places the ink cartridge **126** in communication with the clicker **120**. Similar to the mechanical engagement with the locking mechanism **140**, depression of the clicker **120** may cause a tip **127** of the ink cartridge **126** to extend from the grip portion **114** and a distal end **115** of the dice spinner **200**, such that the dice spinner **200** can be used for writing. In some embodiments, selective depression or manipulation of the clicker **120** (e.g., one depression, two quick depressions, twisting, etc.) may cause the ink cartridge **126** to be deployed independently from the clicker's **120** communication with the locking mechanism **140**. Concomitantly, selective depression or manipulation of the clicker **120** (e.g., one depression, two quick depressions, twisting, etc.) may cause the locking mechanism **140** to be disengaged from the spinner **130** independently from the clicker's **120** deployment of the ink cartridge **126**.

[0049] FIG. **4** illustrates a cross-sectional view of the dice spinner **200** of FIG. **6**. As illustrated, the dice spinner **200** includes the body **110** with a dice portion **112** and a grip portion **114**. The dice portion **112** defines an internal cavity **122** for housing the spinner **130**, the spinner housing **136**, and the locking mechanism **140**. The dice portion **112** also defines a plurality of windows **16** (only a few windows **116** are labelled in FIG. **4**; see also FIG. **1**). Additional windows **16**, may be positioned around the dice portion such that a plurality of numbers, from a single wheel **32**, may be displayed in the separate windows **116**. The grip portion **114** defines a second internal cavity **124** for housing the ink cartridge **26** (e.g., a portion of the ink cartridge **126**). The clicker **120** may be disposed on a top or proximal end **117** of the body **110** and may be in mechanical communication or engagement with the locking mechanism **140**.

[0050] As before, the spinner **130** includes a plurality of wheels **132** each bearing a plurality of numbers **133**. In some embodiments, the plurality of wheels **132** are disposed and spin about the center post **134**. The plurality of wheels **132** may bear sequential, non-sequential, and/or random sets of numbers **133**. By spinning and then stopping, the plurality of wheels **132** will cause a random display of numbers **133** through the plurality of windows **116**, thereby mimicking a dice roll.

[0051] FIG. **10** illustrates the locking mechanism **140** according to the present disclosure. As illustrated, the locking mechanism **140** includes a body **144** having a top end **146** and a bottom end **148**. A plurality of teeth **142** extend between the top and bottom ends **146**, **148**. In some embodiments, each tooth of the plurality of teeth **142** engages (e.g., mechanically engages) one wheel of the plurality of wheels **132** of the spinner **130** (see FIGS. **7-8**) much like, or similar to, engaging the sprocket **34** and wings **35** as set forth previously herein. When the teeth **142** are engaged with the plurality of wheels **132**, movement of the wheels **132**, such as spinning, is prevented. When the teeth **142** are disengaged, the wheels are allowed to move, such as by spinning. Upon re-engagement of the wheels **132** by the teeth **142**, movement of the wheels is stopped. When the wheels **132** have stopped spinning, one of the plurality of numbers **133** on each of the wheels **132** will be visible through each of the plurality of windows **116** (see FIG. **1**). In this way, the dice spinner **200** mimics the rolling or throwing of dice pieces by a player to determine that player's next move, strike, accuracy, etc.

[0052] In some embodiments, the locking mechanism **140** is incorporated as part of the spinner housing **136** (see FIGS. **2-3**). In some embodiments, the locking mechanism **140** is part of the spinner **130**.

[0053] FIGS. **11**A and **11**B illustrate additional embodiments of a dice spinner **200** according to the present disclosure. Specifically, FIGS. **11**A and **11**B illustrate dice spinners, where a body has a worked and ornamented exterior. For example, FIG. **11**A illustrates a dice spinner **300** having a body **310** that includes elements of the spinner **130** (i.e., plurality of wheels **132** each bearing a plurality of numbers **133**). In some embodiments, other elements of the spinner **130** (e.g., the spinner housing) and/or the locking mechanism **140** may still be housed within an internal cavity defined by a body **310**. The clicker **120** may also be worked and ornamental, such as being a character or creature.

[0054] In these alternate embodiments a wheel **332** may, again, be at tension or at rest. For example, referring to FIG. **11**A, the entirety of the wheels **332** may be displayed on an exterior of a pen **300** or other longitudinal member. The wheels **332** may be read down the longitudinal line of the pen at a point specified on the pen, with a notch, arrow, or the like. The wheels **332** may manually spin by a user, such as using an individual's fingers or hands to spin the wheels **332**. The pen **300** may include a clicker and a locking mechanism wherein applying pressure to the clicker disengages the locking mechanism so a user may manually spin the wheels **332**. Once the pressure is no longer applied to the clicker the wheels **332** discontinued spinning and may be "locked" at the specified location. Similar to the disclosure above, one click may release the locking mechanism and another click reengages the locking mechanism. Alternatively, spinning the wheels **332** may include no locking mechanism and the wheels spin until rotational momentum simply comes to halt on its own, without external force from the user.

[0055] As shown in FIG. **11**B, the exterior of the body **410** of the spinner **400** may be worked in the design of an animal or creature-either real or imaginary (e.g., dragons). In some embodiments, the worked design of the body **410** corresponds to the type of game being played using the spinner **200** **100** (e.g., D&D, Magic The Gathering®, Monopoly®, etc.). In some embodiments, the worked design of the body **10** includes gems, metal working, wood working, molded plastic, rhinestones, and other appropriate decorative elements.

[0056] FIGS. **12** and **13** illustrate internal views of another embodiment of a dice spinner according to the present disclosure. Similar to the dice spinner **100** of FIGS. **1-9**, the dice spinner **100'** includes a housing **10'**, where the housing has a top portion **12'** (not illustrated) and a bottom portion **14'**. The top portion **12'** is secured to the bottom portion **14'** via screws, clips, snaps, adhesives, welding, or other appropriate securement mechanism. In some embodiments, the top portion **12'** is removably secured to the bottom portion **14'**. The top portion **12'** of the housing **10'** may define one or more windows **16'** (not illustrated), such as two windows **16** like the dice spinner **100**. In some embodiments, the windows (e.g., windows **16**) allow a user of the dice spinner **100'** to visually see dots, circles, other indicia, etc. corresponding to numerical values. For example, a portion of disks **32'**, or dice disks, are visible through the one or more windows, where the dice disks **32'** carry or bear indicia **38'** corresponding to numerical, or other, values. In this way, the dice spinner **100'** can approximate or mimic a roll of dice during, for example, game play.

[0057] In some embodiments, the dice spinner **100'** additionally includes a crown **20'**, or button. The crown **20'** may be received or disposed within a void **28'** defined by the top and bottom portions **12'**, **14'** of the dice spinner **100'**. In some embodiments, the crown **20'** has a head **22'** seated substantially within the void **18'** and a body **24'** that protrudes through the void **28'** into an interior of the housing **10'**. The head **22'** of the crown **20'** extends a distance from the housing **10'**, such that a user can manipulate the crown **20'** to cause the dice disks **34'** housed within the housing **10'** to spin and display new numerical values. A thickness of the head **22'** may be substantially the same as a thickness of the housing **10'** such that the head **22'** is flush with the housing **10'** when the crown **20'** is manipulated (e.g., when the head **22'** is pressed toward the housing **10'**). The crown **20'** may be "pushed" or "twisted" in a direction to articulate the spinners and/or locking mechanism.

[0058] Similar to the dice spinner **100**, the crown **20'** interfaces or mechanically engages with a locking mechanism, and/or spinning mechanism, housed within the interior of the housing **10'**. The locking mechanism and/or spinning mechanism **40'**, housed within the interior of the housing **10'** of the dice spinner **100'** includes bodies or arms **42'**, stoppers **44'**, and at least one spring **50'**. The arms **42'** may be rigid or semi-rigid arms **42'** incorporated into the body **24'** of the crown **20'**. The arms **42'** are positioned about the body **24'** of the crown **20'**, such that each arm **42'** may engage with a dice disk **32'** positioned within the housing **10'** (e.g., may engage gear **34'** on an underside of the dice disk **32'**, see FIG. **14**B).

[0059] An end **26'**, or distal end, of the crown **20'** may interface and/or mechanically engage the spring **50'**. Disposed at or near the end **26'** are stoppers **44'** for stopping a rotation of the dice disks **32'**. From a top perspective the dice spinner **100'** may have a first side and a second side, or right side and a left side, respectively, which may be mirror images of each other. Complementary, the locking mechanism may have a first side and a second side that mirror each other as well and particularly regarding the arms **42'** and the stoppers **44'**.

[0060] In some embodiments, manipulation of the crown **20'** causes the spring **50'** to compress, thereby causing the arms **42'** and the stoppers **44'** of the locking mechanism **40'** to release from a dice disk **32'**. For example, pressure may be applied to the head **22'** of the crown **20'**, causing the body **24'** and end **26'** of the crown **20'** to extend further into the housing **10'**. Such extension causes the spring **50'** to compress, and the arms **42'** and stoppers **44'** to disengage and/or release the dice disk **32'**. Upon release of the arms **42'** and stoppers **44'**, the dice disk **32'** will spin around a pivot point **36'** (see FIG. **14**A). The arms **42'** may provide the "push" of the dice disks **32'** to force rotation of the dice disks **32'** or alternatively the arms **42'** may disengage the dice disks **32'** wherein the dice disks **32'** may be at tension and release allows for tensioned spin or free spinning of the dice disks **32'**.

[0061] When the crown **20'** is released, the body **24'** and end **26'** of the crown **20'** will retract. Upon release of the crown **20'**, the arms **42'** will re-engage with the dice disks **32'** and the stoppers **44'** will engage a perimeter **37'** of the dice disks **32'**, causing the dice disk **32'** to stop spinning. When the dice disk **32'** stops spinning, indicia **38'** corresponding to a numerical value will be visible through the one or more windows.

[0062] FIGS. **14**A and **14**B illustrate top and bottom views, respectively, of dice disks **32'** for use with the dice spinner **100** of FIGS. **1-9** and/or the dice spinner **100'** of FIGS. **12-13**. The dice disks **32'** include indicia **38'** corresponding to numerical or other values. The dice disks **32'** also include multiple engagement points for engaging with the locking mechanism **40'**. Specifically, the dice disks **32'** include a gear **34'** on a side of the disks **32'** opposite the indicia **38'**. The gear **34'** is for engaging with the arms **42'** of the locking mechanism **40'**. The gear **34'** may incorporate a ball bearing **45'** to facilitate free rotation of the dice disks **32'** upon disengagement of the locking mechanism **40'**. The ball bearing **45'** may correspond substantially to the pivot point **36'**, about which the dice disks **32'** spin.

[0063] The dice disks **32'** also include a perimeter **37'**, which may be a scalloped perimeter, for engaging with the stoppers **44'**. Specifically, the stoppers **44'** may be ball bearings (i.e., rounded) that may engage with impressions of divots in the scalloped perimeter **37'**. In this way, the stoppers **44'** stop rotation of the dice disks **32'**, allowing indicia **38'** to be viewed by a user.

[0064] While particular embodiments have been illustrated and described herein, it should be understood that various other changes and modifications may be made without departing from the spirit and scope of the claimed subject matter. Moreover, although various aspects of the claimed subject matter have been described herein, such aspects need not be utilized in combination.

[0065] In one embodiment, the terms "about" and "approximately" refer to numerical parameters within 10% of the indicated range. The terms "a," "an," "the," and similar referents used in the context of describing the embodiments of the present disclosure (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. Recitation of ranges of values herein is merely intended to serve as a shorthand method of referring individually to each separate value falling within the range. Unless otherwise indicated herein, each individual value is incorporated into the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., "such as") provided herein is intended merely to better illuminate the embodiments of the present disclosure and does not pose a limitation on the scope of the present disclosure. No language in the specification should be construed as indicating any non-claimed element essential to the practice of the embodiments of the present disclosure.

[0066] Groupings of alternative elements or embodiments disclosed herein are not to be construed as limitations. Each group member may be referred to and claimed individually or in any combination with other members of the group or other elements found herein. It is anticipated that one or more members of a group may be included in, or deleted from, a group for reasons of convenience and/or patentability. When any such inclusion or deletion occurs, the specification is deemed to contain the group as modified thus fulfilling the written description of all Markush groups used in the appended claims.

[0067] Any combination of the above-described elements in all possible variations thereof is encompassed by the present disclosure unless otherwise indicated herein or otherwise clearly contradicted by context.

[0068] Although this disclosure provides many specifics, these should not be construed as limiting the scope of any of the claims that follow, but merely as providing illustrations of some embodiments of elements and features of the disclosed subject matter. Other embodiments of the disclosed subject matter, and of their elements and features, may be devised which do not depart from the spirit or scope of any of the claims. Features from different embodiments may be employed in combination. Accordingly, the scope of each claim is limited only by its plain language and the legal equivalents thereto.

What is claimed:

1. A dice spinner comprising:

a housing comprising a top and a bottom, the top defining one or more windows;

a spinner housed within the housing, the spinner comprising a locking mechanism, at least one dice disk, and a spring; and

a crown in mechanical communication with the spinner such that applying pressure to the crown causes the spring to compress, thereby releasing the locking mechanism from the at least one dice disk, allowing the at least one dice disk to spin and display one or more numbers through the one or more windows of the housing.

2. The dice spinner of claim **1**, wherein the locking mechanism comprises an arm for engaging a portion of the at least one dice disk and at least one stopper for engaging a perimeter of the at least one dice disk.

3. The dice spinner of claim **1**, wherein the at least one dice disk comprises a plurality of dice disks.

4. The dice spinner of claim **1**, wherein the at least one dice disk comprises a gear with teeth to engage the locking mechanism, wherein the dice disk comprises indicia corresponding to a number.

5. The dice spinner of claim **4**, wherein the indicia comprises dots.

6. The dice spinner of claim **1**, wherein the crown is in mechanical communication with the locking mechanism of the spinner.

7. The dice spinner of claim **1**, wherein the locking mechanism comprises a body having two sides, each side including a stopper.

8. The dice spinner of claim **7**, wherein the at least one dice disk comprises a plurality of dice disks and each stopper of the locking mechanism is in contact with a perimeter of the respective dice disk, thereby preventing movement of the dice disks.

9. The dice spinner of claim **1**, wherein the top and bottom of the housing are continuous with each other.

10. The dice spinner of claim **1**, wherein the top and bottom of the housing are removably attached to each other.

11. A dice spinner comprising:

a housing comprising a top portion and a bottom portion, the top portion defining one or more windows;

a spinner housed within the housing, the spinner comprising a locking mechanism, a first dice disk in connection with the locking mechanism, and a second dice disk in connection with the locking mechanism; and

a crown received within a void defined between the top portion and bottom portion of the housing, the crown in mechanical communication with the locking mechanism such that applying pressure to the crown causes the locking mechanism to disengage the first and second dice disks, allowing the first and second dice disks to spin.

12. The dice spinner of claim **11**, wherein releasing pressure from the crown causes the locking mechanism to re-engage the first and second dice disks, such that the first and second dice disks stop spinning and display one or more numbers through the one or more windows of the housing.

13. The dice spinner of claim **11**, wherein the crown comprises a head, a body positioned substantially within the housing, and an end opposite the head, the end being in mechanical communication with a spring of the locking mechanism.

14. The dice spinner of claim **13**, wherein the head has a thickness substantially similar to a thickness of the housing.

15. The dice spinner of claim **11**, wherein the locking mechanism comprises a first arm engaged with the first dice disk, a first stopper engaged with a perimeter of the first dice disk, a second arm engaged with the second dice disk, and a second stopper engaged with a perimeter of the second dice disk.

16. The dice spinner of claim **15**, wherein the first stopper and the second stopper comprise ball bearings.

17. The dice spinner of claim **11**, wherein the first dice disk comprises a gear for engaging an arm of the locking mechanism and a perimeter for engaging a stopper of the locking mechanism.

18. The dice spinner of claim **17**, wherein the perimeter of the first dice disk is scalloped.

19. The dice spinner of claim **17**, wherein the second dice disk is identical to the first dice disk.

20. The dice spinner of claim **17**, wherein indicia of the first dice disk corresponds to even numerical values and indicia of the second dice disk corresponds to odd numerical values.

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